



Luvian

AI-Native Systems Engineering

Human-led design. AI-guided modeling.

Seed Round • Confidential • April 2026

THE PROBLEM

MBSE tools are stuck in the last decade.



Expensive

\$5K–\$15K+ per seat per year for enterprise MBSE suites. Incumbent pricing locks out 80% of engineering teams.



Desktop-First

No real-time collaboration. No web access. Install on every machine. Disconnected from the rest of the toolchain.



Zero Intelligence

No AI assistance. No ambient suggestions. Engineers manually trace, validate, and cross-reference everything.

Manual tracing takes **5–10 minutes per requirement** — in programs with thousands of requirements, engineers spend more time on traceability than design. (NASA/SERC 2025)

MARKET OPPORTUNITY

\$4.5B market. Growing 12–16% CAGR.

TAM

\$4.5B

Composite: MBSE + Requirements + Safety + AI

SAM

\$1.8B

Web-native, English-speaking markets

SOM (YEAR 5)

\$30–70M

Category leadership, platform revenue

\$1.8B

MBSE Tools

\$1.2B

Requirements Mgmt

\$0.6B

Safety & V&V

\$0.5B

AI for Engineering

\$0.4B

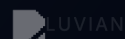
Integration/PLM

Why now: SysML v2 creates a once-in-a-decade tool replacement window

DoD digital engineering mandates

Local LLMs make on-prem AI viable

CONFIDENTIAL / SHARED UNDER NDA / NO SCREENSHOTS / NO RECORDINGS



THE SOLUTION

The Figma of systems engineering, with an **AI partner** built in.

Luvian is a web-native MBSE platform where modeling, requirements, safety analysis, and verification live in one connected graph — with ambient AI that learns from your engineering decisions.



SysML v2 Native

First-class SysML v2 parser, importer, and visual modeling canvas.



Ambient ML

6-stage local-first suggestion engine. AI that never sees your classified data.



Real-Time Collab

Yjs/CRDT-powered multiplayer editing. See your team model in real time.



Deploy Anywhere

Cloud SaaS, on-premises, or air-gapped. One codebase, dual deployment profiles.

One platform. Full coverage.

1 Visual Modeling Canvas

- ✓ Block Definition & Internal Block Diagrams
- ✓ Ports, connectors, nested blocks with auto-layout
- ✓ SysML v2 import & export
- ✓ Interactive canvas with keyboard shortcuts

2 Requirements Engineering

- ✓ INCOSE quality engine (A–F grading)
- ✓ Traceability matrix with coverage analysis
- ✓ Suspect link detection on model changes
- ✓ StrictDoc / ReqIF import

3 Safety Analysis

- ✓ HARA with ISO 26262 ASIL derivation
- ✓ TARA (ISO 21434), STPA, SOTIF, FMEA
- ✓ Hazard-to-requirement linking
- ✓ AI-assisted hazard identification

4 Verification & Validation

- ✓ Test management with evidence attachment
- ✓ Integration readiness reviews
- ✓ Model health validation
- ✓ Coverage engine (7-dimension RCS)

Technical moat.

Ambient ML Pipeline

6-stage local-first suggestion engine: candidate generation, feature extraction, ranking, policy, response shaping, feedback loop. Runs entirely on-device — no data ever leaves the deployment.

Patent-pending

Command → Patch → CRDT

Dispatch pipeline produces forward + inverse patches that flow into Y.Doc for real-time collaboration with undo/redo preservation. Every operation is reversible.

Patent-pending

Dual Deployment Profiles

Commercial (cloud LLM) and government (local-only AI) profiles from a single codebase. Feature flags control the AI boundary policy.

Air-gap ready

Integrated Decision Graph

Architecture, requirements, safety goals, and test evidence in one connected graph. Change a requirement → suspect links propagate to tests, blocks, and hazards.

Single graph, not bolted-on modules

BUSINESS MODEL

SaaS + on-prem. 4 tiers.

Tier	Target	Price / seat / yr	Key Includes
Team	Small teams (5–20)	\$1,200 (\$100/mo)	SysML v2 modeling, collab, requirements, 5 projects
Professional	Mid-market (20–100)	\$2,400 (\$200/mo)	+ Safety analysis, INCOSE quality, ambient ML, API
Enterprise	Large programs (100+)	\$3,600 (\$300/mo)	+ On-prem, SSO/SAML, audit logs, SLA, dedicated support
Defense/Gov	DoD, air-gap (50+)	\$5,400 (\$450/mo)	+ Air-gap AI, IL4/IL5 docs, FedRAMP artifacts, dedicated CSM

70–75%

Subscription Revenue

15–20%

Professional Services

5–8%

AI Compute Add-On

2–5%

Platform / API

3–10x cheaper than incumbent enterprise suites (Cameo, DOORS, Jama). Price-competitive with the new wave at \$100–\$300/editor/mo — differentiation is depth of integration.

Land in engineering. Expand enterprise.

Y1

Design Partners

Founder-led sales. 5 partners. Subsidized pricing for feedback + case studies.

Y2

Early Revenue

First AE. Inbound from content + community. 19 customers. SOC 2.

Y3

Growth

Sales team. Channel partners (Booz Allen, SAIC). FedRAMP. 46 customers.

Y4-5

Scale

Full sales org. Defense enterprise. International. 131 customers.

Target Verticals

- Gov Robotics (DoD Autonomous Systems)
- Software-Defined Vehicles (SDVs)
- Hardware T&V Platforms (dSPACE, Vector, NI)

Deepest integration + local-first AI.

Scope vs. AI Sophistication

Capability Coverage

Capability	Luvian	Flow	Cameo	Trace	Dalus
SysML v2 Modeling	✓	✗	✓	✗	✓
Requirements Mgmt	✓	✓	✗	✓	✓
Safety Analysis	✓	✗	✗	✗	~
V&V / Test Mgmt	✓	~	✗	~	~
Ambient / Local AI	✓	~	~	~	~
Real-Time Collab	✓	✓	✗	✓	✓
Air-Gap / On-Prem	✓	✗	✓	✓	~

Only Luvian integrates all four pillars in one connected graph with ambient local-first AI and CRDT-based real-time collaboration.

Built for classified. Deployable anywhere.

Air-Gap Ready

Single `docker compose up` deploys the full stack. Local-only AI via 80MB MiniLM embeddings. Zero data egress.

Docker

ONNX Runtime

Local LLM

Standards Literacy

DO-178C, DO-254, ARP4754A/4761, ISO 26262, ISO 21434 built into safety analysis and V&V workflows.

Avionics

Automotive

Defense

Compliance Path

SOC 2 Type 1 targeted post-seed. FedRAMP Li-SaaS planned Y3. IL4/IL5 authorization path mapped for Y4.

SOC 2

FedRAMP

IL4/IL5

\$270K+

Min defense ACV (50 seats)

\$5M+

Enterprise-wide deal potential

\$15M+

Max defense ACV (500+ seats)

Built by engineers who live in the domain.

F

Founder & CEO

Full-stack architect. Built the 3-layer rendering pipeline, ambient ML engine, and defense deployment architecture. Domain expertise in systems engineering tooling.

SW

SW Engineer

20-year veteran. Production hardening: CI/CD, security, observability, performance. Enterprise infrastructure at scale.

PM

Product Manager

Product launch and market fit. Personas, GTM strategy, analytics, onboarding, and beta program execution.

Post-Funding Hiring Plan (Y1)

- Head of Sales
- Senior Frontend
- Ons / G&A
- Solutions Engineer
- Backend / ML Engineer

Path to \$21M revenue by Year 5.

Revenue & Costs (Base Case)



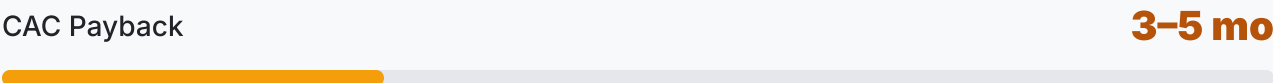
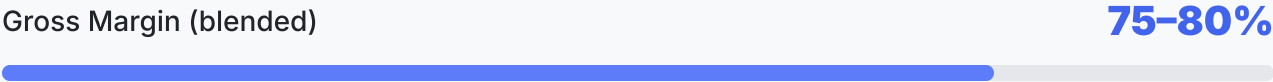
Revenue Scenarios



Base Case	Y1	Y2	Y3	Y4	Y5
Revenue	\$165K	\$846K	\$3.3M	\$9.9M	\$21.2M
Costs	\$1.7M	\$3.7M	\$7.2M	\$11.8M	\$16.8M
Net Income	(\$1.6M)	(\$2.9M)	(\$3.9M)	(\$1.9M)	\$4.4M

Best-in-class LTV:CAC.

LTV:CAC Ratio by Segment (target: >3x)



Benchmark: \$300-500K for best-in-class SaaS

THE ASK

Raising \$7–10M Seed

24+ months runway to product-market fit and first \$1M ARR.

Build the Team

4 eng, 1 product, 1 SE, 1 sales, 1 ops. Hire the team that scales past the founder.

Close First Revenue

5 design partners. SOC 2 Type 1. First enterprise contracts. \$100K+ ARR by month 12.

Establish the Category

Define "AI-native MBSE" before incumbents respond. File patents. Build advisory board.

Funding Milestones

Seed (now)

\$7–10M

PMF signal: 3+ paying, NPS > 40

Series A (Y2)

\$20–30M

\$1M+ ARR, 120%+ NRR

Series B (Y3–4)

\$50–80M

\$5M+ ARR, defense contracts



Luvian

Human-led design. AI-guided modeling.

The platform for safety-critical systems engineering.

Market	\$4.5B TAM, 15-16% CAGR
Model	\$100-\$450/seat/mo SaaS + on-prem
Raise	\$7-10M seed, 24+ months runway
Moat	Integrated decision graph + local-first AI

Luvian Confidential — NDA Restrictions